

## Technical Report: Implementation of Bio-Sand Filtration

**Location:** Daw Nye Khu IDPs Camp in Huai Pu Long, Karenni State

**Implementing Agency:** CTER – Karenni

**Reporting Period:** May 18 – 23, 2026

---

### Executive Summary

The implementation of the bio-sand filtration tank in Huai Pu Long for Daw Nye Khu IDP camp and middle school students was successfully carried out from May 18 to 23, 2026, with the active participation of local residents.

### Socio-Political Climate

Following the military coup in 2021, Huai Pu Long village became one of the conflict-affected communities, forcing the entire village population to flee to the Thai-Karenni border for safety. Although residents have now returned to their village, many displaced people from Daw Nye Khu village in Demoso Township remain in Huai Pu Long and are still unable to return home.

Despite the ongoing challenges, the education system has continued to operate as normally as possible. However, both displaced persons and students have been relying on untreated water directly from nearby streams for daily use. The water quality becomes especially poor during the rainy season and after rainfall, posing serious health and sanitation concerns.

### Existing Water Infrastructure Challenges

The school and displaced communities relied on a water system that faced several systemic challenges, including:

- **Mineral Contamination:** The water contained excessive limestone and mineral deposits, making it unsuitable for safe drinking.
- **Infrastructure Damage:** During the rainy season, flooding frequently damaged the water pipelines, disrupting access to water for both students and displaced residents.

### Project Methodology and Activities

- Conducted a community meeting to discuss the work plan and ensure the effective implementation and timely completion of the planned activities.
- Renovated the water pipeline system from the water source to the village.
- Installed a bio-sand filtration system to improve access to safe and clean drinking water.

## Community Mobilization

Implementation began on May 18, 2026, under the supervision of two hired water specialists to oversee the work plan and construction process. Community members actively contributed volunteer labor to construct the tank and collect sand, stones, and timber required for the project.

## Filtration and Storage Installation

The project utilized a multi-stage Bio-Sand filtration approach:

- **Filtration System:** Two 2,000-liter, one 1,000-liter, and one 500-liter tank were installed.
- **Layering Technique:** The primary tank utilized a graded media filter consisting of large/medium/small stones, charcoal, and rough/fine sand. Subsequent tanks utilized similar layers without charcoal.
- **Foundation:** A cement-based platform was constructed to support the filtration units.

## Pipeline Renovation

Some parts of the water pipeline were replaced, and LDPE water pipes were installed at the water source to reduce damage caused by strong water flow during heavy rains. In addition, a small water storage tank was installed at the water source to improve water storage capacity and prevent debris and waste materials flowing into the pipeline.

## Beneficiary Demographics

The project directly serves a displaced and returning population:

- **Total Population:** 112 individuals (52 male; 60 female)
- **Educational Support:** 56 students (30 male; 26 female)

## Infrastructure Improvements

- **Accessibility:** An additional water outlet from the storage tank was installed to supply water to the toilets, improving access to sanitation facilities.
- **Better Water Quality:** The filtered water now provides improved water quality for drinking and cooking purposes.

## Challenges and Sustainability

Despite the project's success, several external constraints were identified:

- **Maintenance Concerns:** Strong water flow during the rainy season frequently damages and breaks the water pipelines. The community often faces difficulties in repairing the system due to limited financial resources to replace damaged pipes.
- **Capacity Building:** To ensure sustainability, community members received technical knowledge through the installation of the bio-sand filtration system.